

Ujaan Das

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EDUCATION

University of Warwick

Master of Science in Computer Science

Hong Kong University of Science and Technology

Bachelor of Engineering in Computer Engineering

Coventry, United Kingdom

Sep. 2025 – Sep. 2026

Hong Kong SAR

Sep. 2021 – May 2025

EXPERIENCE

Software Engineering Intern (Part Time)

Stellerus Technology

Jan. 2025 – Aug. 2025

Hong Kong SAR

- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%
- Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions

Software Engineering Intern

Stellerus Technology

May 2023 – Aug. 2023

Hong Kong SAR

- Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

Research Assistant

HCI Initiative, HKUST

Jun. 2022 – Aug. 2022

Hong Kong SAR

- Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments
- Collaborated with post-docs to build a modular React + TypeScript frontend using custom hooks and Tailwind CSS, improving component reusability and streamlining development across multiple researchers
- Contributed to FARPLS, a feature-augmented preference learning system that improved label quality and user engagement without increasing cognitive load (published at ACM/IUI'24)

PROJECTS

MSc Dissertation - Invariants | C++, Gtest, CMake, Python

[Github](#)

- Applying formal systems theory to LLMs for improved structured output generation via runtime constraints
- Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines

Snip | C++, Termios, Gtest, CMake, Nix

[Github](#)

- Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture

TerraViz | Next.js, React, TypeScript, CesiumJS, TailwindCSS

[DevPost](#)

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data

oojsite | Go, HTML, TailwindCSS, Nix

[Github](#)

- Built a static site generator in Go to compile Markdown into composable HTML, actively used for [my blog](#)

SKILLS

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: GTest, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic, Celery

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git

AWARDS

HackEurope'26 Scholar, Published to IEEE ICSE'27, Cathay Hackathon 2025 Finalist, Published to ACM IUI'24, Cathay Hackathon 2024 Winner, Study Abroad Scholarship in the USA, Undergraduate Entry Scholarship